

Pearls AQI Predictor: An End-to-End Machine Learning System for Three-Day Air Quality Forecasting in Islamabad

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Abstract—Air pollution is a significant public-health concern in Islamabad, where particulate-matter concentrations vary sharply with weather and season. This work presents Pearls AQI Predictor, an end-to-end machine learning system that forecasts the Air Quality Index (AQI) for the next 24, 48, and 72 hours. Two years of hourly air-quality and meteorological data are obtained from the Open-Meteo API and transformed into lag, rolling, wind-vector, cyclical, and interaction features. Three multi-output regression models—Random Forest, XGBoost, and LightGBM—are trained to predict future fine particulate matter (PM_{2.5}), which is then mapped to the US EPA AQI. The best model attains a coefficient of determination (R^2) of 0.73 at the 24-hour horizon, decreasing to 0.64 at 72 hours. The trained model is deployed through an interactive Streamlit dashboard, and scheduled GitHub Actions automate hourly feature updates and daily retraining. The system demonstrates a complete MLOps pipeline from data ingestion to live, automated deployment.

Index Terms—air quality index, PM_{2.5}, time-series forecasting, machine learning, gradient boosting, MLOps, Streamlit

I. INTRODUCTION

Exposure to fine particulate matter (PM_{2.5}) is associated with respiratory and cardiovascular illness. Islamabad experiences pronounced seasonal pollution, particularly during winter when low wind speeds trap pollutants near the surface. Reliable short-term forecasting of the Air Quality Index (AQI) enables residents and authorities to take timely precautions.

This paper describes Pearls AQI Predictor, a system that forecasts AQI for the next three days using only freely available data. The contributions are: (i) an automated feature pipeline built on the Open-Meteo API requiring no API key; (ii) a comparative study of three multi-output regression models for multi-horizon PM_{2.5} prediction; (iii) an interactive web dashboard for live forecasts and health alerts; and (iv) a fully automated retraining pipeline implemented with GitHub Actions.

II. RELATED WORK

AQI forecasting has been approached with statistical models such as ARIMA, classical machine learning methods such as Random Forests and gradient-boosted trees, and deep sequence models such as LSTMs. Tree-based ensembles are widely used because they handle non-linear interactions between pollutants and weather without extensive preprocessing. Comparable student and industry projects commonly couple a feature store

with a model registry and schedule periodic retraining. The present work adopts this architecture using lightweight, no-cost components so that the entire pipeline can run on free tiers.

III. METHODOLOGY

A. Data Acquisition

Hourly air-quality and weather data for Islamabad (latitude 33.68°N, longitude 73.05°E) are retrieved from the Open-Meteo air-quality and historical-weather endpoints. The air-quality response provides PM_{2.5}, PM₁₀, carbon monoxide, nitrogen dioxide, sulphur dioxide, and ozone. The weather response provides temperature, relative humidity, wind speed, wind direction, precipitation, and surface pressure. Two years of history are merged on the timestamp, yielding approximately 17,500 hourly records. Fig. 1 shows the PM_{2.5} time series and its right-skewed distribution.

B. AQI Computation

The US EPA AQI is computed from PM_{2.5} concentration C using piecewise-linear interpolation:

$$\text{AQI} = \frac{I_{hi} - I_{lo}}{C_{hi} - C_{lo}} (C - C_{lo}) + I_{lo}, \quad (1)$$

where (C_{lo}, C_{hi}) and (I_{lo}, I_{hi}) are the concentration and index breakpoints of the band containing C .

C. Feature Engineering

A total of 44 features are derived, grouped as follows:

- **Log transform:** $\log(1 + \text{PM}_{2.5})$ to stabilise variance against extreme spikes.
- **Lag features:** PM_{2.5} at 1, 2, 3, 6, 12, 24, and 48 hours in the past.
- **Rolling statistics:** mean and standard deviation over 3, 6, 12, and 24-hour windows.
- **Pollutant lags:** PM₁₀, NO₂, and CO at 1 and 6 hours.
- **Wind vectors:** $w_x = v \cos \theta$, $w_y = v \sin \theta$ from wind speed v and direction θ .
- **Cyclical time:** sine and cosine encodings of hour and month.
- **Interaction features:** a stagnation index $\log(1 + \text{PM}_{2.5})/(v+1)$, a weekend flag, and 24-hour temperature and humidity tendencies.

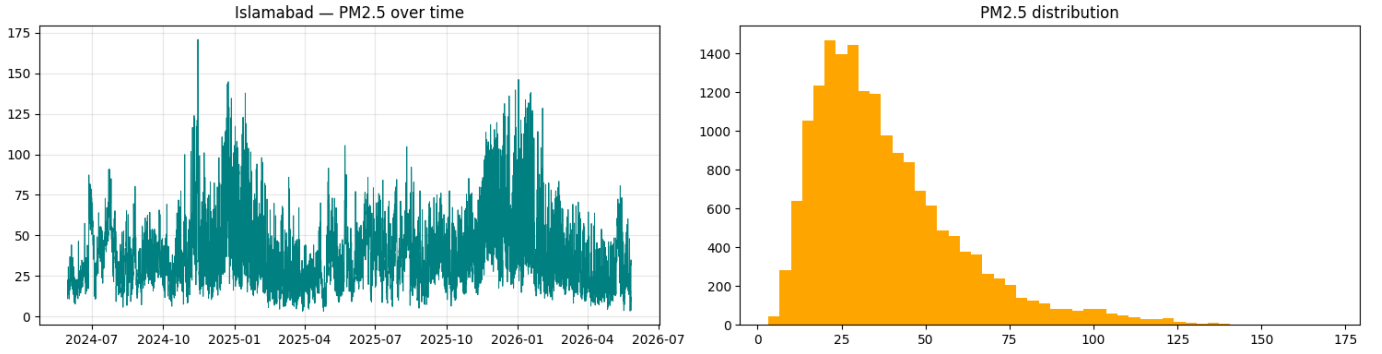


Fig. 1. Exploratory data analysis: PM_{2.5} over two years (left) and its right-skewed distribution (right).

The prediction targets are the raw PM_{2.5} values 24, 48, and 72 hours into the future.

D. Model Training

Three multi-output regressors are trained, each predicting all three horizons simultaneously: Random Forest, XGBoost, and LightGBM. A chronological 80/20 train-validation split is used so that the validation set always follows the training set in time, which is appropriate for time-series data. Models are evaluated with Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The model with the highest average R^2 is selected; for deployment, the compact LightGBM model is preferred because it achieves comparable accuracy with a far smaller artifact.

E. Automation Pipeline

Two scripts are scheduled with GitHub Actions. The feature pipeline runs hourly, fetching recent data, recomputing features, and updating a versioned feature store (`feature_store.csv`). The training pipeline runs daily, retraining on the full history and committing the updated model (`aqi_models.pkl`). Because the dashboard reads the model from the same repository, retraining automatically refreshes the live application.

F. Dashboard

An interactive dashboard is implemented with Streamlit and deployed on Streamlit Community Cloud. It loads the trained model, fetches live data, displays current conditions, the three-day forecast with colour-coded categories, health alerts, and a recent-trend chart. The dashboard is publicly available at <https://aqi-dashboard-iqynzk7oxdupfvvpjjbrhj.streamlit.app/>.

IV. RESULTS AND DISCUSSION

Table I reports the validation R^2 for each model and horizon. The three models perform similarly, with Random Forest achieving the highest average R^2 . Table II presents the detailed error metrics for the best model.

Fig. 2 compares predicted and actual PM_{2.5} on the validation set. The 24-hour forecast is the most accurate, and accuracy declines with the forecast distance. This pattern is expected: as

TABLE I
VALIDATION R^2 BY MODEL AND HORIZON

Model	24h	48h	72h
Random Forest	0.706	0.662	0.643
LightGBM	0.705	0.618	0.590
XGBoost	0.702	0.609	0.609

TABLE II
DETAILED METRICS FOR THE BEST MODEL (RANDOM FOREST)

Horizon	MAE	R^2
24h	8.74	0.706
48h	9.83	0.622
72h	10.20	0.602

the horizon lengthens, current conditions carry progressively less information about future pollution. SHAP analysis of the 24-hour model indicates that recent PM_{2.5} lags and rolling means are the dominant predictors, followed by wind and the stagnation index, which is consistent with the physical mechanism by which low wind speeds trap pollutants.

Compared with coastal cities, Islamabad’s pollution profile is more volatile, with sharp winter spikes that are intrinsically harder to predict. The achieved accuracy is therefore competitive for an inland location with limited free data.

V. CONCLUSION

This work presented Pearls AQI Predictor, a complete and automated machine learning system for three-day AQI forecasting in Islamabad. Using two years of free Open-Meteo data and engineered time, lag, rolling, wind, and interaction features, multi-output tree-based models predict PM_{2.5} which is mapped to the US EPA AQI. The best model reaches $R^2 = 0.73$ at 24 hours and 0.64 at 72 hours. The trained model is served through a live Streamlit dashboard, and scheduled GitHub Actions keep the feature store and model continuously up to date. The project demonstrates an end-to-end MLOps workflow built entirely on free-tier services.

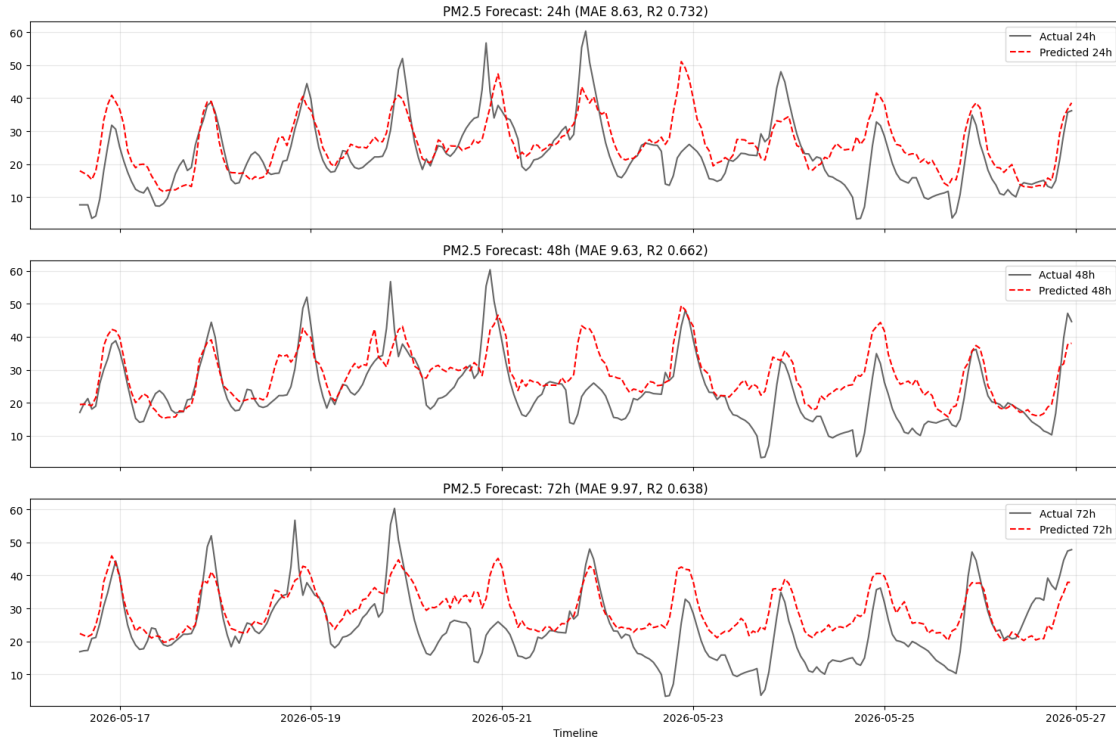


Fig. 2. Actual versus predicted $PM_{2.5}$ on the validation set for the 24h, 48h, and 72h horizons.

VI. FUTURE WORK

Future directions include migrating the feature store to a managed cloud database, incorporating weather forecast data to improve the 48 and 72-hour horizons, adding sequence models such as LSTMs for comparison, extending the system to multiple cities, and delivering email or SMS alerts when hazardous AQI is forecast.

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